Dineth Wijesinghe

Software Engineer Intern

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SUMMARY

Information Technology undergraduate at the University of Moratuwa with a strong foundation in software engineering, full-stack application development, Clean Architecture, and distributed systems. Hands-on experience developing robust backend services, high-throughput APIs in Go and Node.js, and modern React web applications, with experience deploying scalable containerized solutions on AWS. Focused on applying software engineering principles to build clean, maintainable, test-driven, and reliable systems; seeking a Software Engineering internship.

EDUCATION

BSc (Hons) in Information Technology **2024 – Present**

University of Moratuwa, Moratuwa, Sri Lanka

- **GPA :- 3.6 / 4.0**

G.C.E. Advanced Level **2020 – 2022**

H/North Kanumuldeniya National School, Hambantota, Sri Lanka

- Combined Mathematics | Physics | ICT · **Z-Score :- 1.7991**

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, C#, SQL, Bash
Backend Development: Node.js, Express.js, Bun, Hono, FastAPI, RESTful APIs, WebSockets
Frontend Development: React, React 18, Vite, Tailwind CSS, HTML5
Databases & ORM: PostgreSQL, MySQL, SQLite, MongoDB, Redis, DynamoDB, Entity Framework Core
Cloud & DevOps: Docker, Kubernetes, Terraform, Jenkins, AWS, CI/CD, Linux, Git, GitHub

KEY PROJECTS

WarehousePOS – Desktop Point of Sale & Warehouse Management System GitHub Repo

▶ **Technologies :-** C#, .NET 10, WPF, MVVM, EF Core, SQLite, Dependency Injection, Moq, xUnit, Serilog

- Designed a modular monolith desktop application with strict Clean Architecture separation across Domain, Application, Infrastructure, and UI layers.
- Implemented robust transactional state machines for purchase orders and sales checkout, guaranteeing ACID compliance and zero unrecorded inventory adjustments.
- Established a high code-quality standard with zero-warning compilation (`TreatWarningsAsErrors`), automated unit/integration tests with xUnit & Moq, and single-file self-contained deployment.
- Managed schema evolution and data persistence using EF Core with SQLite, configuring comprehensive audit logs and daily disaster-recovery database backups.

CloudSight – Cloud Observability Platform (Group Project) GitHub Repo

▶ **Technologies :-** Node.js, Express.js, React, Python, scikit-learn, Docker, AWS (EC2, RDS, ECR, Amplify), PostgreSQL, Redis, Nginx, CI/CD

- Contributed to a collaborative full-stack cloud observability platform for proactive monitoring, anomaly tracking, and incident management of web applications.
- Developed the automated anomaly detection process, threshold-based incident tracking workflows, and real-time WebSocket streaming with multi-tier severity classification.
- Architected an automated zero-downtime CI/CD delivery pipeline using GitHub Actions, Amazon ECR, and Docker Compose on EC2 behind an Nginx reverse proxy with private Amazon RDS.

Tunnix – Self-Hosted Developer Tunneling & Ingress Platform GitHub Repo

▶ **Technologies :-** Go, gorilla/websocket, Bun, Hono, React 18, Vite, Docker, AWS (EC2, CloudFront, S3), Nginx, SQLite (WAL), TypeScript

- Developed a high-performance Go ingress gateway multiplexing thousands of concurrent HTTP requests over bi-directional WebSocket frames with sub-50ms latency and cross-platform CLI agents.
- Architected a Bun/Hono control plane with SQLite in WAL mode, a 4-tier token security model, and a real-time React 18 dashboard for live tunnel telemetry and custom subdomain management.
- Automated production deployment and edge routing leveraging AWS EC2, CloudFront, S3, dynamic Nginx reverse proxying, and automated Let's Encrypt SSL certificates.

ShopSphere – Containerized Microservices Platform on AWS GitHub Repo

▶ **Technologies :-** Amazon EKS, Kubernetes, AWS ALB Ingress Controller, Terraform, Docker, Multi-AZ VPC, RDS PostgreSQL, DynamoDB, SNS, SQS, HPA

- Provisioned an enterprise microservices platform on Amazon EKS across a Multi-AZ VPC with public/private/database subnets, NAT Gateways, and path-based routing via the AWS Load Balancer Controller.
- Automated infrastructure provisioning using modular Terraform, configured Horizontal Pod Autoscaling (HPA) via Metrics Server, and decoupled order/payment services using SNS/SQS queues with Dead-Letter Queues.

AWS K3s OpenTelemetry Observability Lab GitHub Repo

▶ **Technologies :-** AWS EC2, K3s Kubernetes, OpenTelemetry Collector, Honeycomb, OTLP/HTTPS, Docker, Linux, Bash

- Provisioned a production-grade K3s cluster on AWS EC2 running a 30+ microservices system with AWS Security Group network isolation, namespace CPU/memory resource quotas, and self-healing health probes.
- Engineered a unified OpenTelemetry Collector pipeline exporting telemetry over secure OTLP to Honeycomb, monitoring high-cardinality p95/p99 latency waterfall traces to eliminate performance bottlenecks.

UniNet – Cross-Platform Network Automation Utility GitHub Repo

▶ **Technologies :-** Bash, PowerShell, Linux NetworkManager Dispatcher, macOS launchd, Windows Task Scheduler, curl

- Engineered a zero-dependency, cross-platform utility automating university Wi-Fi captive portal authentication across Cisco/NEC, Aruba Networks, and Ruijie controller environments.
- Implemented event-driven background OS triggers (Linux NetworkManager dispatcher, Windows WLAN-AutoConfig Event 8001 hook) with auto-dependency healing and 1-line curl/PowerShell web installers.

CERTIFICATIONS & COURSES

- Supervised Machine Learning: Regression and Classification – DeepLearning.AI & Stanford Online
- AWS Cloud Essentials – Amazon Web Services · IAM, VPC, EC2, RDS, and Cloud Architecture *(Ongoing)*
- LFS158: Introduction to Kubernetes – The Linux Foundation · Architecture, Deployments & Pod Networking *(Ongoing)*
- LFS101: Introduction to Linux – The Linux Foundation · Linux Administration, Shell Scripting & System Processes
- CCNA (Cisco Certified Network Associate) – Enterprise Networking, Routing & Switching, TCP/IP *(Ongoing)*

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **Batch Representative** | **Faculty of Information Technology, University of Moratuwa** 2026 – Present
- **Programme Committee Member** | **Riseup Mora 2026, University of Moratuwa** 2026
- **Senior Prefect** | **H/North Kanumuldeniya National School** 2019 – 2020

REFERENCES

Mr. Indika Wimalasuriya
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